KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

COLLEGE OF ENGINEERING

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

TECHNICAL REPORT ON STUDENT INTERNSHIP PROGRAMME UNDERTAKEN AT TI MICROELECTRONICS LAB

BY

POBI KWAKU GABRIEL

INDEX NUMBER – 1859722

FROM

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ACKNOWLEDGEMENT

I express my deepest gratitude to the Almighty God for his unsullied grace and blessing for seeing me through. My appreciation goes to the institution, Prof. K. O. Boateng, and his graduate assistant students, Chris and Nero Novo, and his teaching assistants, Barnabas and Matthaeus, for the wonderful opportunity to work alongside them. I want to say a big thank you to all of them, as well as my co-partner, Master Frank Aidoo, for his contributions and collaboration throughout our internship project. Finally, my other colleague interns, who made the whole period worthwhile, I hope we get the opportunity to meet once again and learn as a group.

ABSTRACT

As mandated by the Kwame Nkrumah University of Science and Technology, in preparation for the final year of study in selected undergraduate engineering programs, students are to undertake a practical industrial attachment.

The initiative seeks to supplement the amassed theoretical knowledge acquired by the student from the years of study with a consolidated practical exposure to industrial operations closely related to the particular field of engineering, in order to suitably equip the future engineer with the foundation to meet the ever-growing field demands of the industries and the world at large.

This report documents an eight-week vacation industrial attachment focused on digital systems design and error detection and correction in hardware design. This experience effectively bridged the gap between abstract modern algebra and practical hardware implementation in computer hardware design.

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CHAPTER ONE

1.1 INTRODUCTION AND BACKGROUND

The basic initiative for this attachment is to provide a link between the theoretical and practical sides of learning, and to give students the necessary exposure to their various fields of work. As part of my curriculum and to fulfill the requirements for my four years of study at the university, I completed the third year in the Electrical and Electronic Engineering department.

I spent my 8-week practical training at the Computer Engineering Lab (TI Microelectronics Lab).

1.2 OBJECTIVES OF THE INTERNSHIP PROGRAM

- The attachment aims to enable students to relate their theoretical knowledge to field work.
- Exposing students to the world of work associated with their field of study.
- Exposing students to work places challenges and its solutions.
- Allows students to gain exposure, develop creativity, maintain a good working ethic, build links and build strong network with persons of great experience in the field.

1.3 OVERVIEW OF THE TI MICROELECTRONICS LABORATORY

The TI Microelectronics Laboratory, situated within the computer engineering department at Kwame Nkrumah University of Science and Technology (KNUST), functions as an electronics facility. It is outfitted to facilitate hands-on learning and research in microelectronics, digital system design, and very-large-scale integration (VLSI) studies. The lab is equipped with National Instruments (NI) ELVIS platforms, FPGA (Field-Programmable Gate Array) development boards, and a range of electronic testing and measurement tools essential for circuit implementation, validation, and troubleshooting. The NI ELVIS systems offer a unified platform for circuit prototyping, signal generation, oscilloscope measurements, and data acquisition. FPGA development boards facilitate hardware description language (HDL) implementation and verification of digital systems, allowing students to synthesize and test designs on actual hardware. The laboratory also contains necessary test instruments, including

oscilloscopes, function generators, power supplies, and multimeters, which aid in circuit-level experimentation and validation. This facility supports academic and project work in electronics, microprocessor systems, digital logic implementation, and VLSI design practices. Through hands-on use of these tools and platforms, students gain practical experience in hardware modelling, testing, debugging, and system-level verification.

CHAPTER TWO

DETAILED ACCOUNT OF WORK PERFORMED

2.1 Week 1

Fundamentals of Digital System Design

The internship commenced with a comprehensive revision of Digital Systems Design concepts, specifically focusing on the hierarchical separation of digital systems into datapath and controller design –the main blocks of any digital system. I was introduced to Hardware Description Languages (HDL), focusing on System Verilog fundamentals. During this period, I implemented various digital logic building blocks, including comparators, adders, logic gates, multiplexers (muxes), decoders (demuxes), flip-flops, and counters. This foundational week was essential for establishing the coding standards and logical reasoning required for more complex designs

2.2 Week 2

Sequential Logic and the max_max2 Algorithm

During the second week, I focused on simulating various basic digital building blocks using Xilinx Vivado. I designed and implemented both combinational and sequential components using System Verilog (HDL). An important activity this week was the initiation of a mini-project that required developing an algorithm to calculate and display the maximum and second maximum values among three inputs (*x, y, and z*). I used several combinations of three numbers to be able to generate a C pseudo-algorithm for the project. I then drafted the initial datapath and control unit logic, focusing on the conditional comparisons needed for the algorithm.

ALGORITHM

```
input x,y,z
If (x>y):
        max = x
        max_2 = y
else:
        max = y
        max_2 = x
If (z>largest):
        max_2 = max
        max = z
else:
        (if z>max_2):
                max_2 = z
display max, max_2
```

FIGURE 1: THE ALGORITHM FOR CALCULATING AND DISPLAYING THE MAXIMUM AND SECOND MAXIMUM NUMBER (MAX_MAX2 PROJECT)

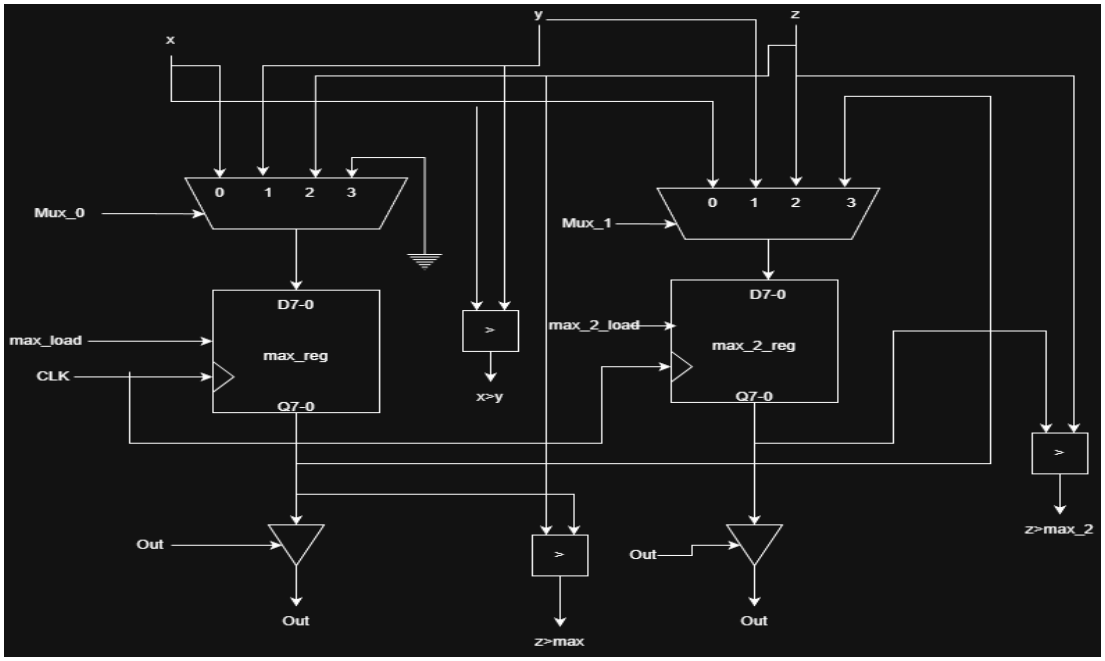

FIGURE 2: DATAPATH FOR THE MAX_MAX2

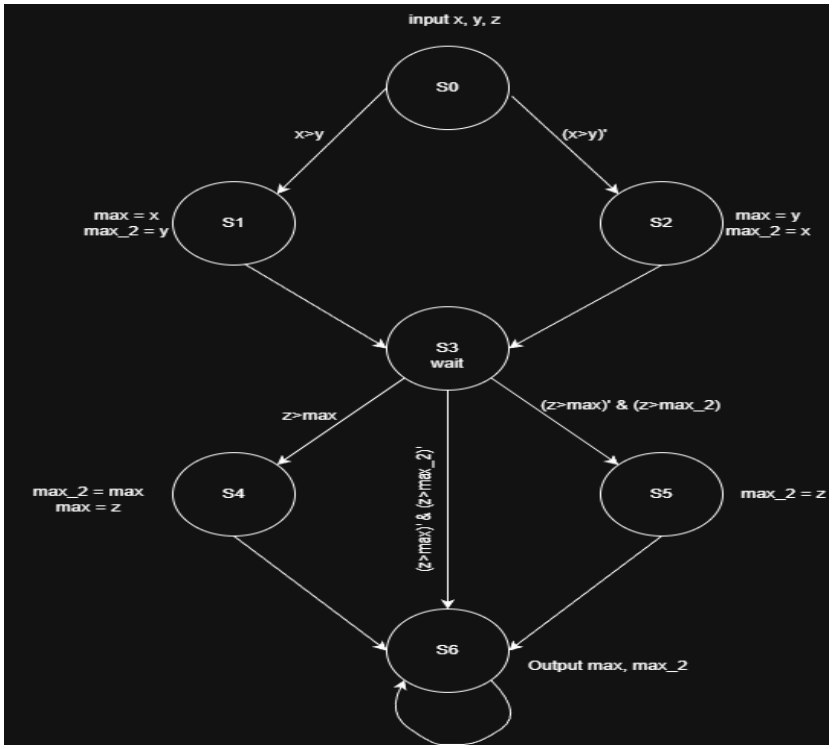

FIGURE 3: CONTROLLER FOR MAX_MAX2

2.3 Week 3

Project Completion and Advanced Digital Applications

This week, I moved on to the programming and simulation of the max_max2 datapath and controller in Xilinx Vivado using SystemVerilog. I developed a comprehensive test bench to verify the algorithm across different input sets. I then presented my design of the algorithm together with the testbench on the max_max2 project. I moved on to designing and simulating memory arrays and systolic arrays using System Verilog. Finally, I was introduced to cryptography, studied basic ciphers such as Playfair and Caesar, and began studying Galois Fields, which are crucial for error-correcting codes.

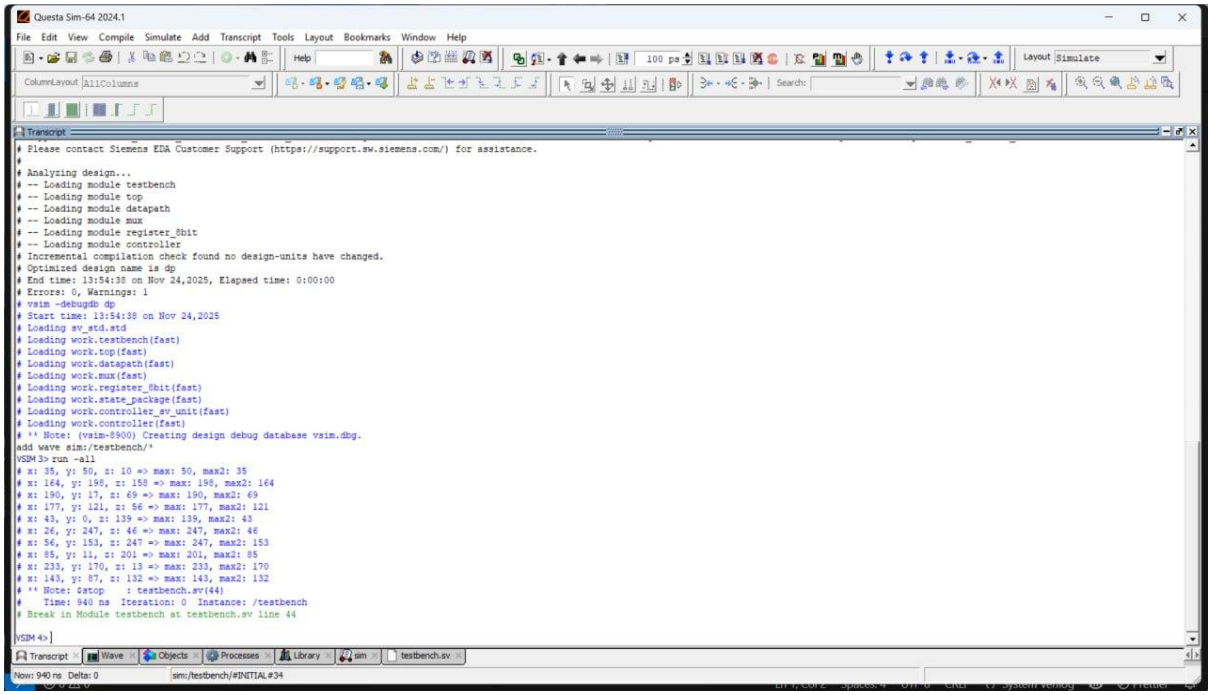

FIGURE 4: TESTBENCH RESULTS FOR MAX_MAX2

2.4 Week 4

Modern Algebra for Error-Correcting Codes

During the fourth week, I selected the digital system applications we were introduced to – namely, error detection and correction. I was given some materials on error coding, which I used to start studying the prerequisites for my project. I learnt modern algebra, including the properties of Groups and Finite Fields, which are the fundamentals of error coding. I then moved on to other concepts, such as the construction of Galois Fields GF (2^m) and the representation of field elements as polynomials. I ended the week with a presentation on vector spaces and matrices over finite fields.

2.5 Week 5

Characterization of Linear Block Codes

This week, I continued working through the materials on the structural characterization of Linear Block Codes. I learnt how an (n, k) linear block code is defined by its generator matrix (G) and parity-check matrix (H). I further learnt that the n-k bits added to a message are called redundant or parity-check bits, which provide the code's error-detecting and correcting capabilities. I focused mainly on concepts such as systematic encoding, syndrome calculation for error detection ($sr\cdot$), and the relationship between a code's minimum distance, and its error-correcting capability.

2.6 Week 6

Binary Cyclic Codes and Systematic Encoding

In the sixth week, I studied the fundamentals and structural properties of Binary Cyclic Codes. I learnt that these codes are an important subclass of linear block codes because they can be efficiently implemented using Linear Feedback Shift-Registers (LFSRs). I studied how to calculate the generator polynomial $g(X)$ and the process of systematic encoding, which involves dividing a message polynomial by $g(X)$ to find the parity-check bits. The remainder after the division is done in GF (2), including Meggitt decoding and error-trapping for Hamming and quadratic residue codes (QR codes).

2.7 Week 7

BCH Code Theory and Iterative Decoding

During my seventh week, I began studying binary Bose-Chaudhuri-Hocquenghem (BCH) codes – a special type of binary cyclic code, specifically the primitive and non-primitive varieties. I learnt the procedures for syndrome calculation, used to detect errors, and the Berlekamp-Massey (BM) iterative algorithm to find the error-location polynomial from the calculated syndromes. Furthermore, I studied Chien’s search method, which is used to find the exact bit locations of errors in a received codeword. On the last day of that week, I presented on key concepts such as syndrome calculation, the BM algorithm, and Chien’s search to locate the error location in binary BCH codes to decode them.

2.8 Week 8: Implementation of (31, 11, 5) BCH Code

In the final week, I used my own example of a t-correcting BCH code – a (31, 11) 5-error-correcting BCH code- to design the hardware using the knowledge gained from the concepts I learnt during my sixth and seventh weeks. This involved calculating the generator polynomial for a code that corrects 5 errors in a 31-bit codeword. I designed the encoder hardware in SystemVerilog and verified its functionality using Questa Sim. I chose an arbitrary message codeword and then used it to calculate the expected codeword using the generator matrix. I then wrote a testbench to verify the design's software implementation against the calculated codeword to be transmitted over the channel. I finally presented on the encoder I designed.

2.8.1 METHODOLOGY

- First of all, I took an example of BCH codes that is capable of detecting and correcting a maximum of 5 errors in received polynomial; thus a (31, 11) 5-error correcting capability BCH code.
- I then found the primitive polynomials for this example, which are:m_1,m_3,m_5 and m_7 from the primitive elements of the GF (2^m).

  m_1= X^5+X^2+1

  m_3= X^5+X^4+X^3 + X^2+1

  m_5= X^5+X^4+X^2+X+1

  m_7=X^5+X^3+X^2+X+1
- The generator matrix is the LCM $\{m_1, m_3, m_5, m_7\}$and it is calculated as the product of the minimal polynomials.
- The calculated generator matrix is X^{20}+X^{18}+X^{17}+X^{13}+X^{10}+X^9+X^7+X^6+X^4+X^2+1
- So the coefficients of the generator matrix form the connections to the Linear Finite Shift Registers which are typically the registers used to hold the values of the parity check bits.
- The number of parity-check bits is given as n-k which the number of codeword bits – the number of message bits: n-k = 31-11= 20. So the number of flip flops needed to form the LFSRs are 20 starting from b_0 –b_{19}.
- I then chose an arbitrary 11-bit message tuple to be encoded into a codeword using the encoding circuit. The message tuple was (11011000001).

- I performed hand calculations using the encoder circuit to determine the LFSR states after 20 micro-operations, to determine the expected parity-check bits to be concatenated with the message bits to form the codeword for transmission over the channel, using the encoding circuit I designed with the generator matrix specifying the connections between the LFSRs.

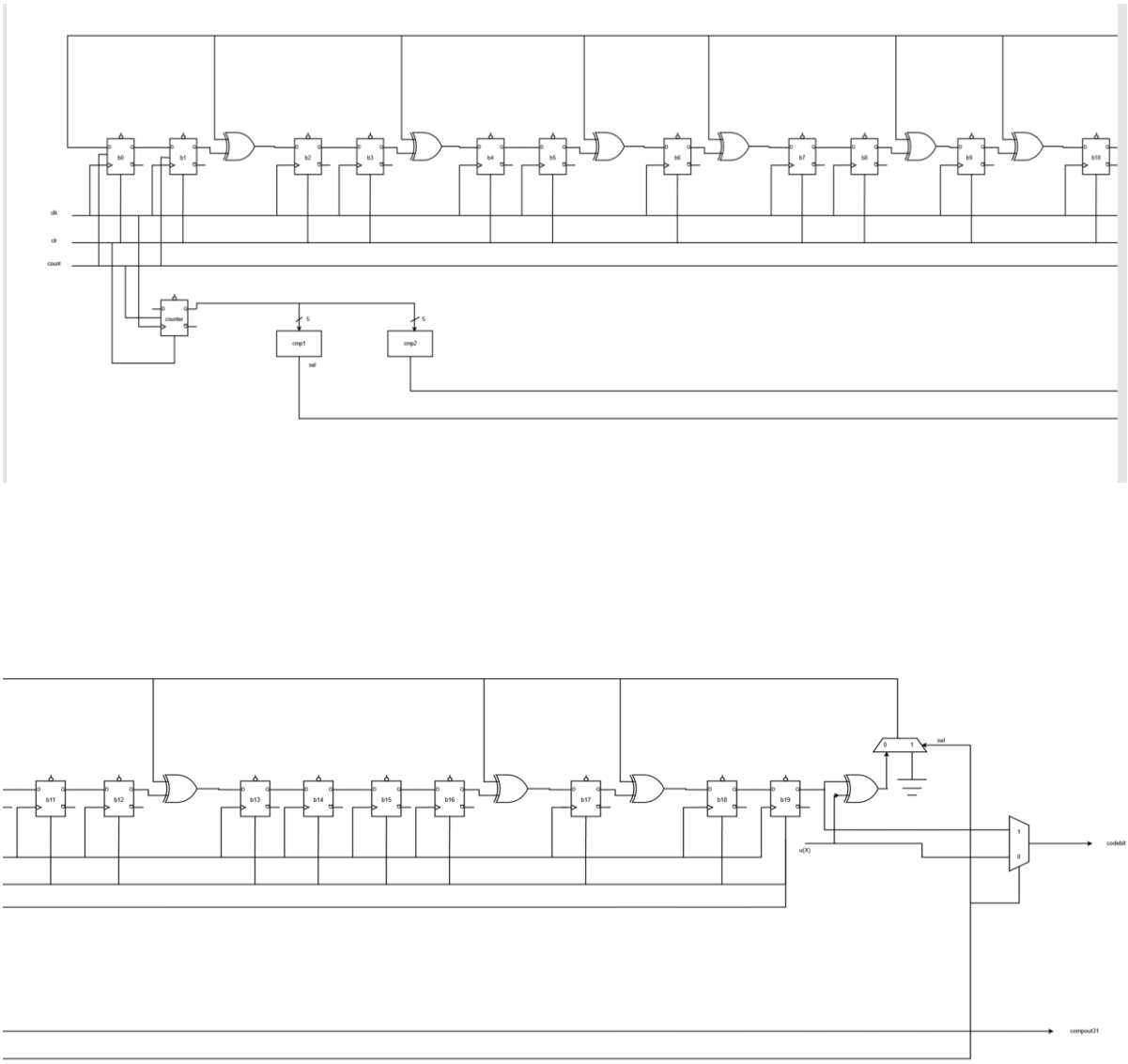

FIGURE 5: THE DESIGNED DATAPATH FOR ENCODING CIRCUIT FOR THE (31, 11, AND 5) BCH CODE

- The table below shows the computation of the parity-check bits for this BCH code using the encoding circuit designed from the (31, 11, 5) BCH code.

u_n	b_0	b_1	b_2	b_3	b_4	b_5	b_6	b_7	b_8	b_9	b_{10}	b_{11}	b_{12}	b_{13}	b_{14}	b_{15}	b_{16}	b_{17}	b_{18}	b_{19}
1	1	0	1	0	1	0	1	1	0	1	1	0	0	1	0	0	0	1	1	0
0	0	1	0	1	0	1	0	1	1	0	1	1	0	0	1	0	0	0	1	1
0	1	0	0	0	0	0	0	1	1	0	1	1	1	1	0	1	0	1	1	1
0	1	1	1	0	1	0	1	1	1	0	1	1	1	0	1	0	1	1	0	1
0	1	1	0	1	1	1	1	0	1	0	1	1	1	0	0	1	0	0	0	0
0	0	1	1	0	1	1	1	1	0	1	0	1	1	1	0	0	1	0	0	0
1	1	0	0	1	1	1	0	0	1	1	0	1	1	0	1	0	0	0	1	0
1	1	1	1	0	0	1	0	1	0	0	0	0	0	0	0	1	0	1	1	1
0	1	1	0	1	1	0	0	1	1	1	1	0	0	1	0	0	1	1	0	1
1	0	1	1	0	1	1	0	0	1	1	1	1	0	0	1	0	0	1	1	0
1	1	0	0	1	1	1	0	1	0	0	0	1	1	1	0	1	0	1	0	1

FIGURE 6: A TABLE SHOWING THE STATES OF THE LFSRS FOR THE ENCODING CIRCUIT

Therefore, the codeword is then given as (1001110100011101010111011000001) and is transmitted over the channel.

2.8.2 RESULTS

I then simulated the designed encoding circuit on Questa Sim in system Verilog to compare hand calculated results to simulation results using testbench. I also generated the schematic of the components I used in designing the encoding circuit such as the datapath and the controller. Below are simulated results

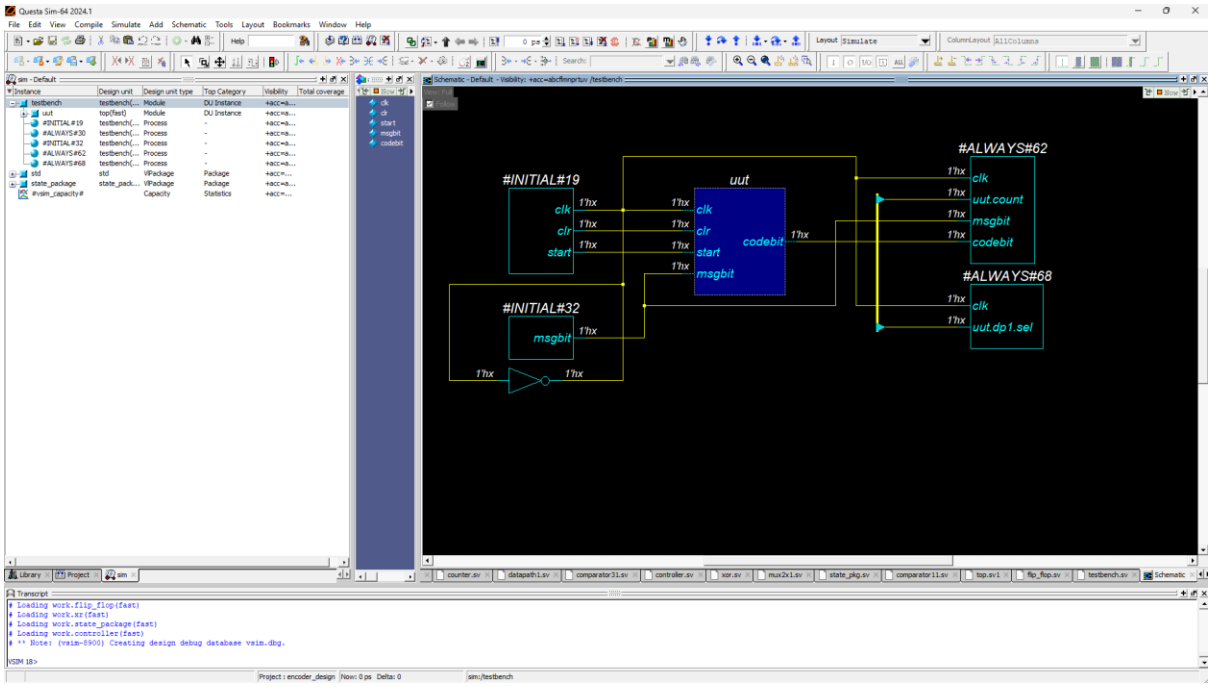

FIGURE 7: THE TESTBENCH SHOWING THE UNIT UNDER TEST (UUT) THUS THE ENCODING CIRCUIT

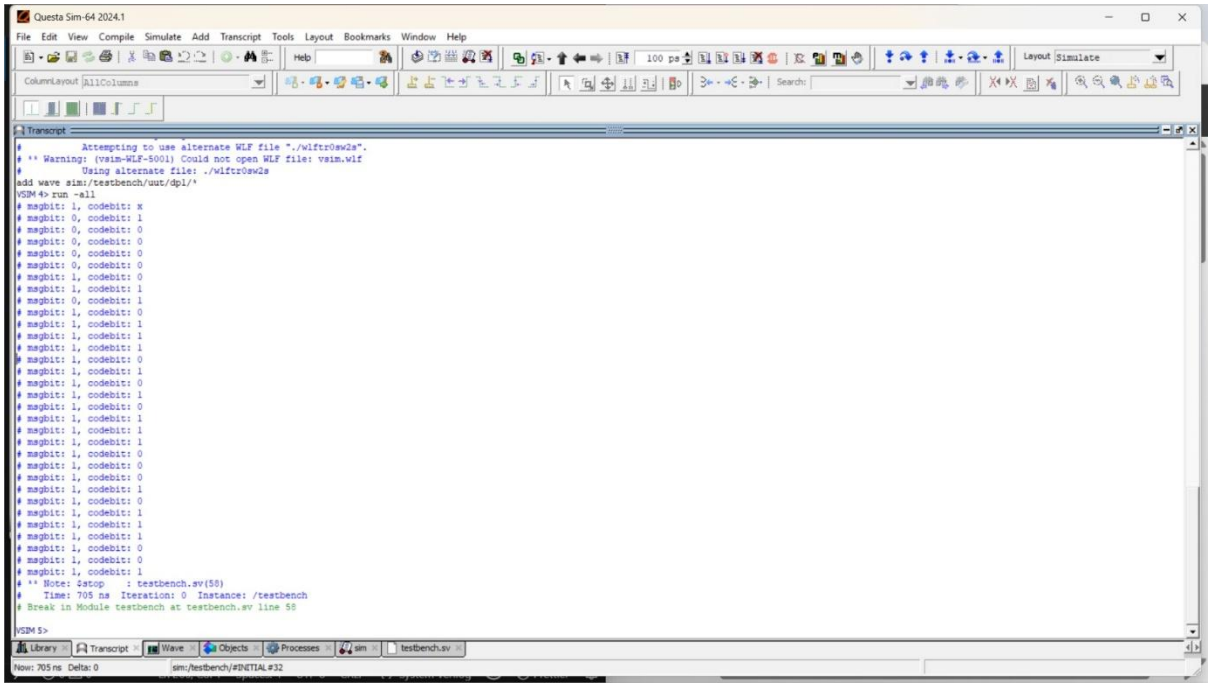

FIGURE 8: TESTBENCH RESULTS FOR THE ENCODING CIRCUIT OF THE BCH CODE

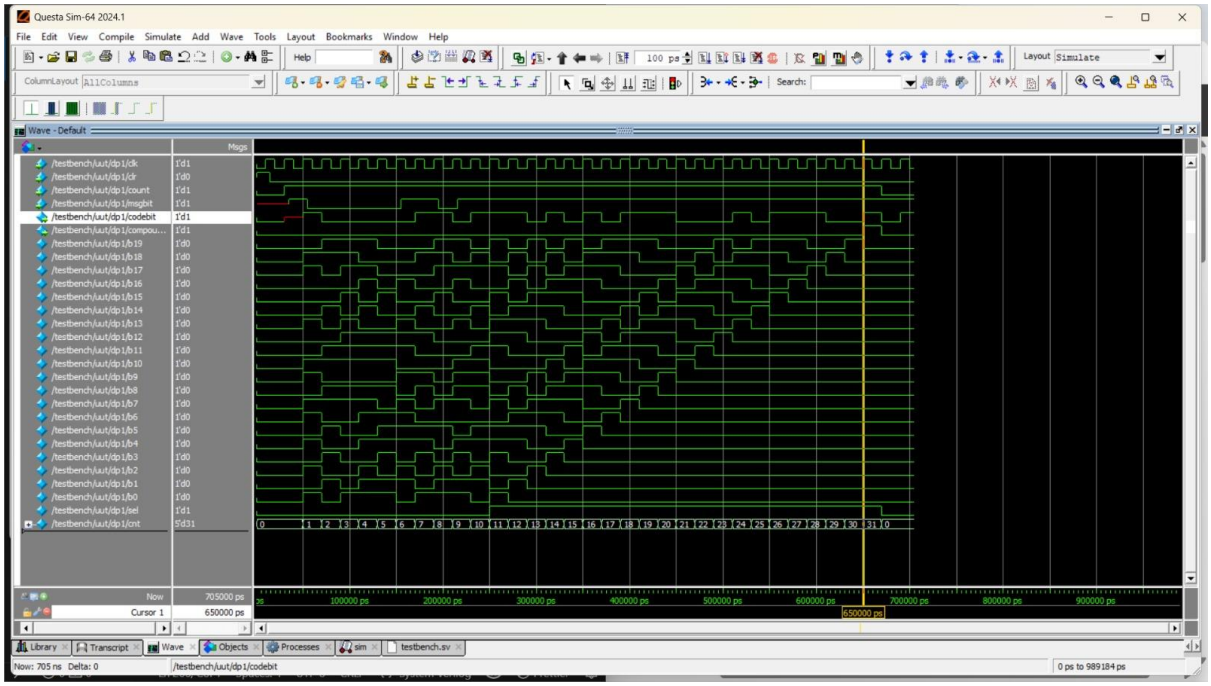

FIGURE 9: WAVEFORM OF THE SIMULATED TESTBENCH

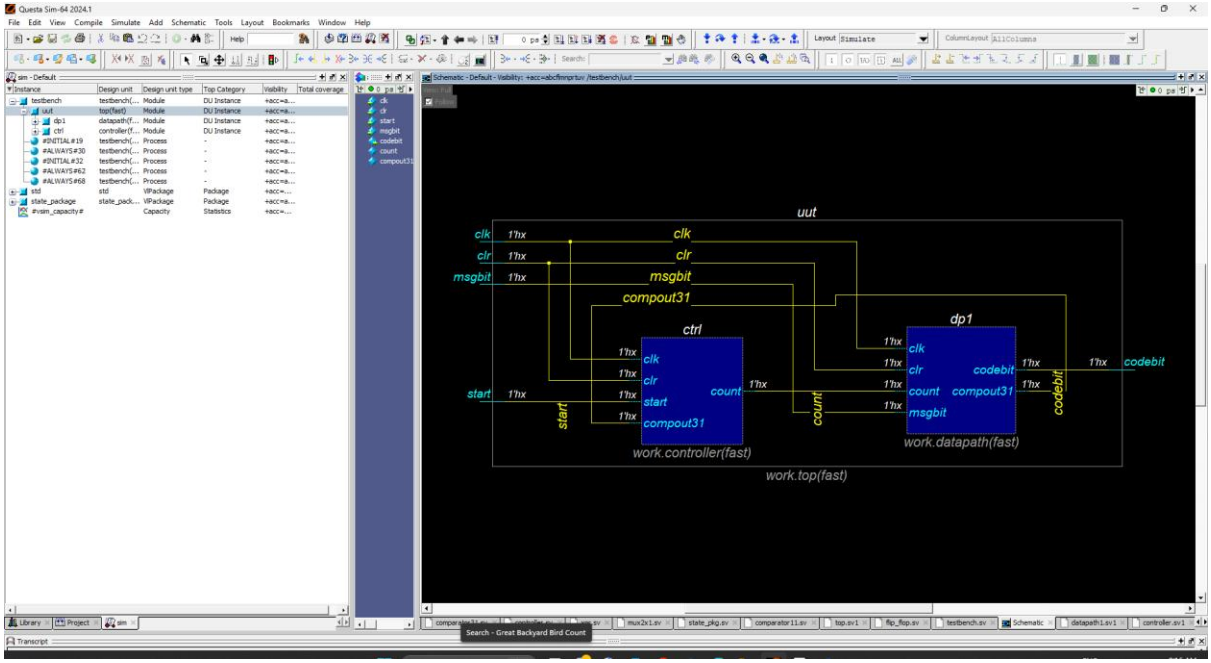

FIGURE 10: THE SIMULATED ENCODING CIRCUIT (UUT)

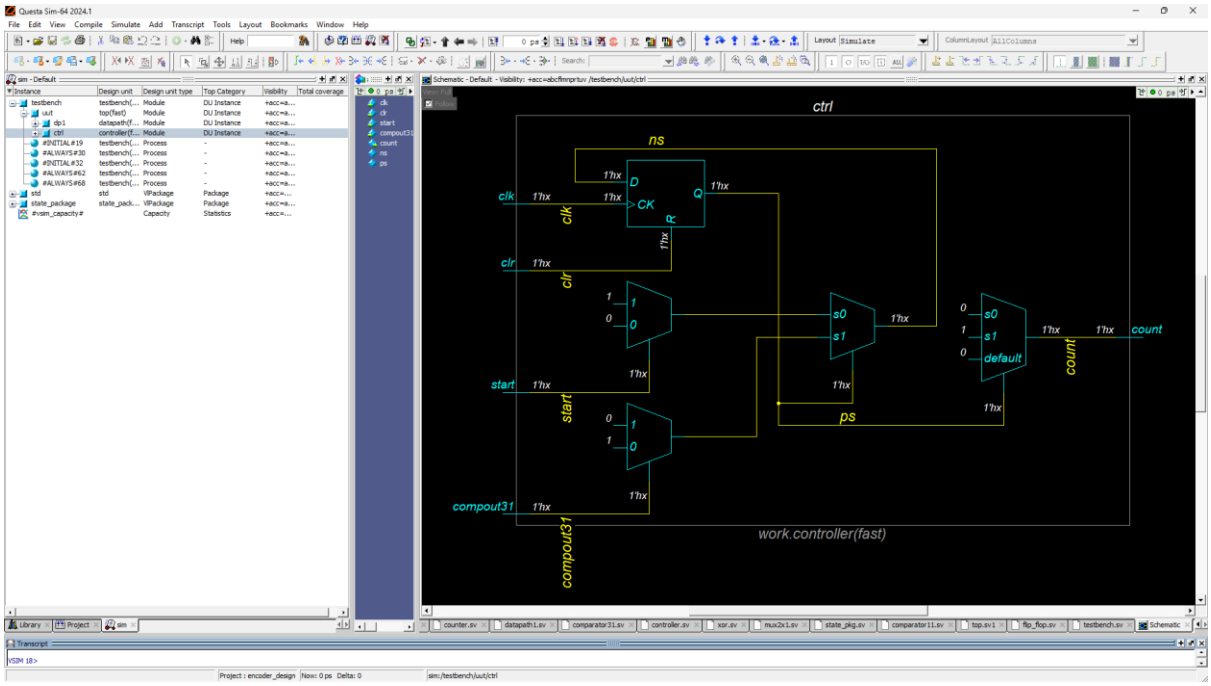

FIGURE 11: THE CONTROLLER FOR THE ENCODING CIRCUIT

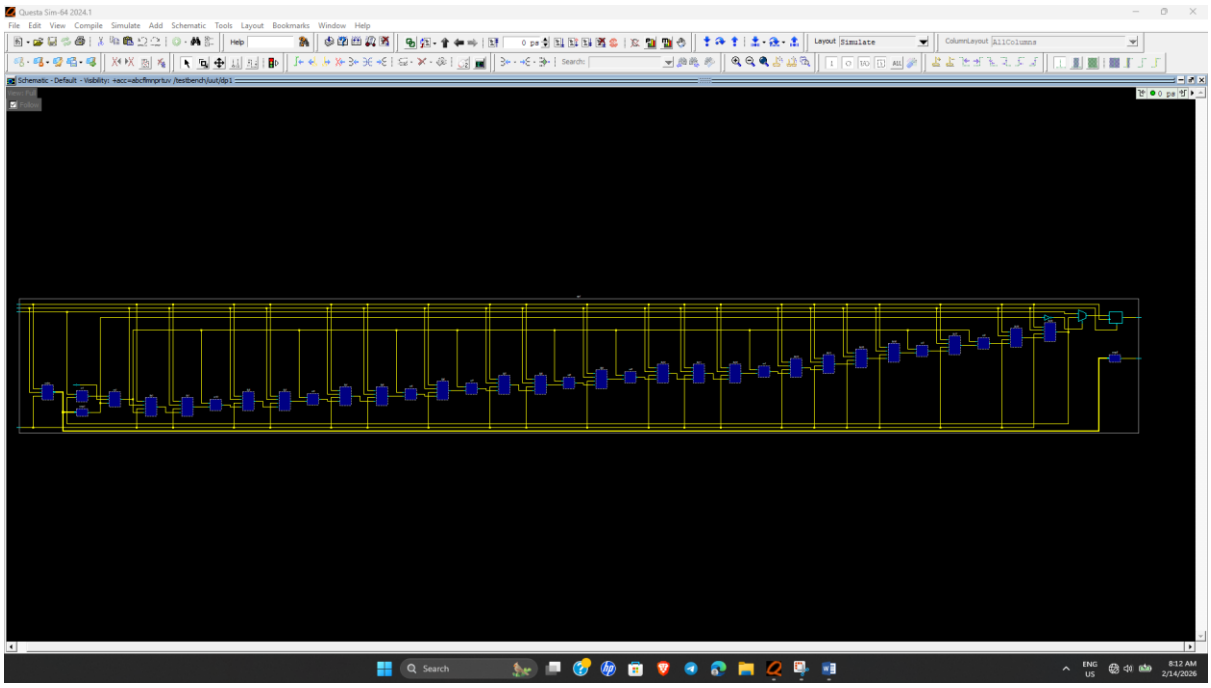

FIGURE 12: THE DATAPATH FOR THE SIMULATED ENCODING CIRCUIT

Below are the clearer view of the datapath for the encoding circuit for the (31, 11, 5) BCH code.

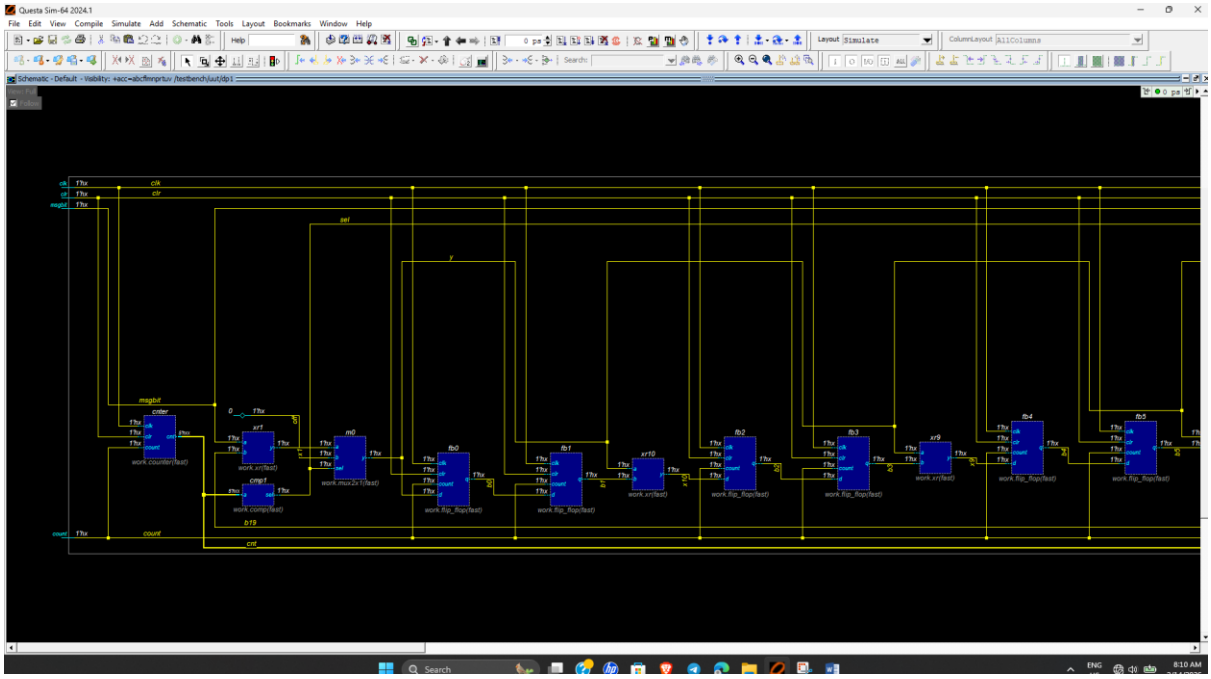

FIGURE 13

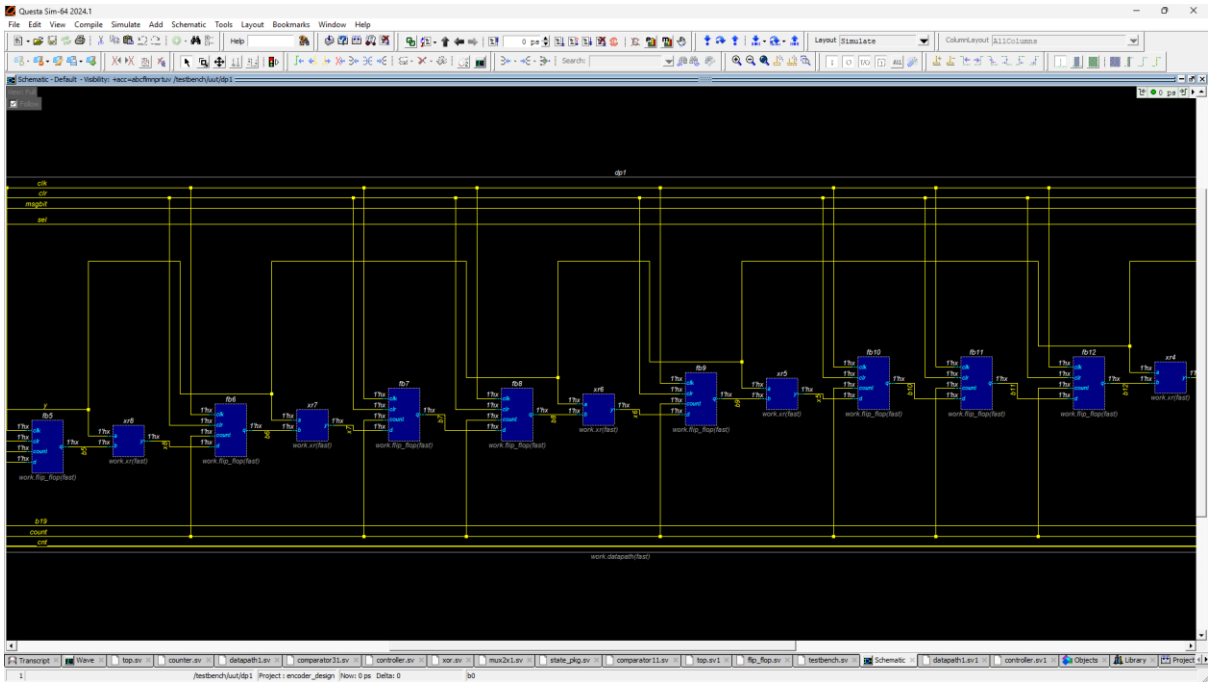

FIGURE 14

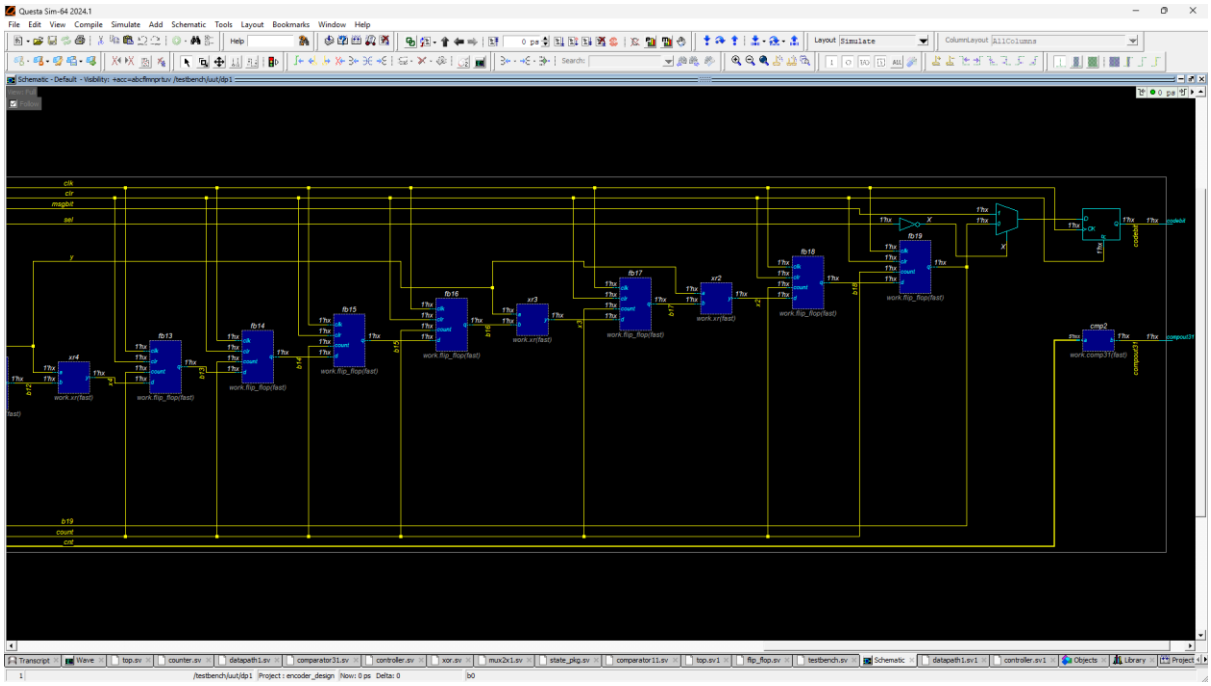

FIGURE 15

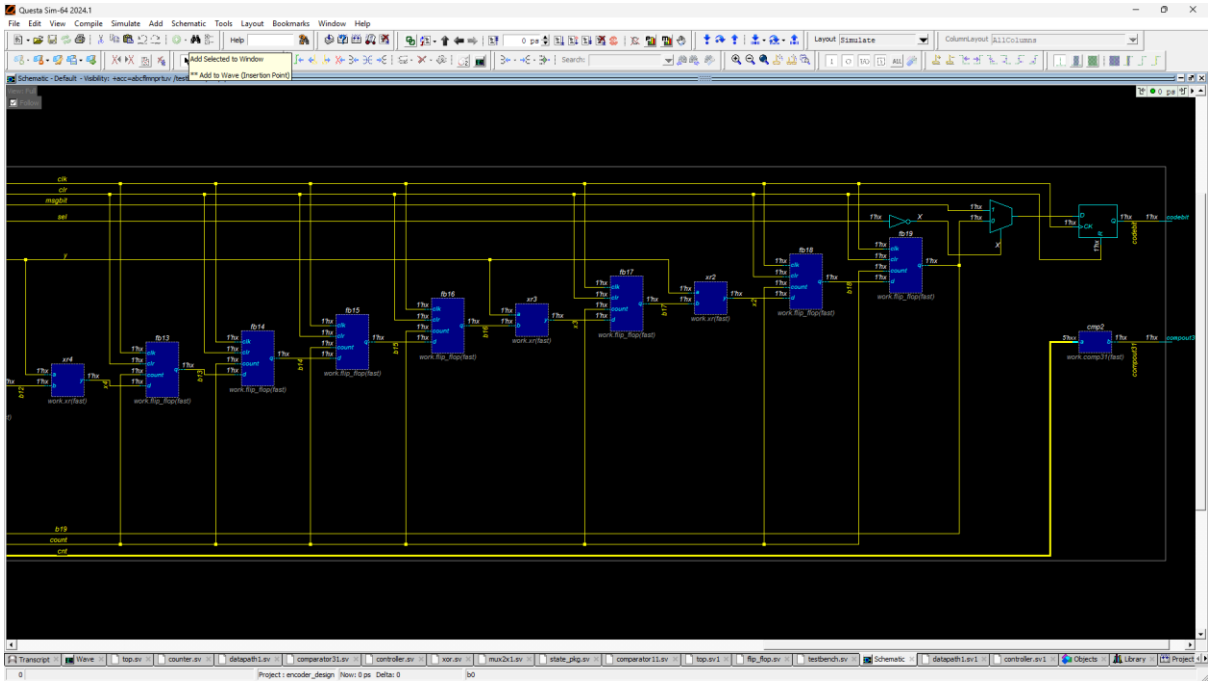

FIGURE 16

CHAPTER THREE

RESULTS AND OBSERVATIONS

3.1 Results and observations for the max_max2 design

The simulation results obtained for the max_max2 datapath and controller verified the validity of the algorithm for calculating and displaying the maximum and second maximum values among the three inputs (x), (y), and (z).

From the testbench results, it was observed that for all applied input combinations, the circuit correctly displayed the largest value as *max* and the second largest value as *max2*. This confirms that the conditional comparison sequence implemented in the controller correctly follows the intended algorithm.

The finite state machine (FSM) successfully coordinated the comparison and data transfer operations in the datapath. Each state activated the appropriate control signals to select the correct operands for the comparator and to update the internal registers holding the current maximum and second maximum values.

I also observed that separating the design into a datapath and a controller significantly simplified debugging and verification (synthesis and testbench). Errors related to incorrect value updates can be traced to either incorrect control signals or to datapath connections, without modifying the entire design.

The waveform results show that the outputs change only at valid clock edges, indicating that the design is fully synchronous and free from unintended combinational glitches at the outputs.

Another important observation is that the design properly handles different input orderings. That is, the correctness of the result does not depend on the position of the largest or second-largest value among the three inputs. This confirms that the comparison logic is independent of input arrangement.

The simulation further demonstrated that the number of clock cycles required to obtain valid outputs is fixed and determined by the controller's state count. This shows that the algorithm was successfully transformed into a predictable hardware execution sequence.

Overall, the max_max2 project demonstrated how a simple conditional C-pseudo-style algorithm can be systematically mapped to a hardware architecture using a structured datapath and a finite-state controller.

3.2 Results and observations for the (31, 11, 5) BCH encoding circuit

The encoding circuit for the (31, 11, 5) BCH code was successfully implemented and verified through simulation. The results from the testbench matched the manually calculated parity bits and final codeword.

For the selected 11-bit message input, the simulated output codeword exactly corresponded to the hand-calculated codeword. This confirms the correctness of the generator polynomial implementation and the overall encoder structure that I designed.

The waveform results show that the Linear Feedback Shift Register (LFSR) stages update correctly at every clock cycle, as determined by the feedback connections derived from the generator polynomial. This verifies that the XOR feedback network accurately implements polynomial division in (GF (2)).

It was observed that after the complete sequence of message bits is shifted into the encoder, the contents of the LFSR registers represent the correct remainder of the polynomial division process. These register values form the parity-check bits that are appended to the message to generate the final codeword.

The controller successfully managed the sequencing of operations, including loading the message bits, enabling the feedback paths, and controlling the register shift operations. This confirms that the control logic correctly synchronizes the datapath during encoding.

The simulation also showed that the number of clock cycles required to complete encoding is directly related to the number of message bits and the number of parity registers. This reflects the theoretical requirement that systematic BCH encoding is performed by serially processing the message through the LFSR structure.

From the schematic and datapath views, it was observed that the encoder architecture is highly modular and scalable. By changing the generator polynomial and the number of registers, the same design structure can be adapted for other BCH code configurations.

Another important observation is that the hardware implementation clearly demonstrates the relationship between abstract Galois field polynomial operations and physical digital hardware. Polynomial arithmetic is implemented directly using only flip-flops and XOR gates.

Finally, the close agreement between the manual calculations and the simulation results confirms the reliability of the encoder design and validates the correctness of the mathematical model used for the BCH code implementation.

CHAPTER FOUR

4.1 CONCLUSION AND RECOMMENDATIONS

This industrial training successfully strengthened my understanding of digital system design and its application to real hardware problems. Through the design and simulation of the *max_max2* system, I gained practical experience in translating algorithmic decision logic into a structured datapath and controller architecture using finite state machines. This helped me clearly understand how control signals coordinate data manipulation and operations in digital systems.

The implementation of the (31, 11, 5) BCH encoder further deepened my understanding of error detection and correction techniques and their realization in hardware. By designing the encoding circuit using Linear Feedback Shift Registers and verifying the results with simulation, I was able to establish a direct connection between Galois field theory and practical digital hardware implementation. The agreement between the hand calculations and simulation results confirmed the correctness of the design and strengthened my confidence in hardware verification using testbenches.

Overall, the internship provided valuable hands-on experience in System Verilog, digital architecture design, simulation, and verification with the help of Xilinx Vivado and Questa Sim. It also bridged the gap between theoretical concepts in communication and modern algebra and their real-world application in computer and communication hardware systems. This training has prepared me with a strong foundation for future academic and professional work in digital systems and communication engineering.

Finally, I would recommend that future interns and students undertaking similar digital systems design and coding theory projects are encouraged to first develop a strong foundation in datapath and controller–based design before attempting more complex systems such as error-correcting code hardware, cryptography and many others. A clear understanding of finite state machines and control signal sequencing greatly simplifies both design and debugging.

4.2 REFERENCES

1. **KNUST College of Engineering**, *Guidelines for Accessing Students' Vacation Training Reports*, 2025.

2. **S. Lin and J. Li**, *Fundamentals of Classical and Modern Error-Correcting Codes*, Cambridge University Press, 2022.

3. **P. K. Gabriel**, *Vacation Training Log Sheets (Weeks 1-8)*, KNUST, 2025.

4.3 APPENDICES

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Our Ref: CoE-VACT/2024 Date:

2024/ 2025 Vacation Training Log Sheet

Name of Student: POBI KWAKU GABRIEL. Index No: 1859122
Department: ELECTRICAL/ ELECTRONICS ENG
Company: II MICROELECTRONICS LAB
Training Period: From OCTOBER 1, 2025 to NOVEMBER 21, 2025
Week Begins: OCTOBER 2, 2025 Week Ends: OCTOBER 3, 2025 Week No: 1

DATE	ACTIVITIES UNDERTAKEN
10/01/2025	Revision of Digital Systems Design concepts; datapath design and controller design.
10/02/2025	Introduction to Hardware Description Language; system Verilog fundamentals.
10/03/2025	Implementation of modules for digital logic building in system verilog like comparators, adders, logic gates, muxes, decoder, flipflops, counters

Name of Supervisor: NERO NOVOR
Title: LAB TECHNICIAN

Date: 13TH FEBRUARY, 2026 Signature:

COLLEGE OF ENGINEERING

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Our Ref: CoE-VACT/2024 Date:

2024/ 2025 Vacation Training Log Sheet

Name of Student: POBI KWAKU GABRIEL Index No: 1859722
Department: ELECTRICAL/ELECTRONICS ENG
Company: II MICROELECTRONICS LAB
Training Period: From OCTOBER 1, 2025 to NOVEMBER 21, 2025
Week Begins: OCTOBER 6, 2025 Week Ends: OCTOBER 10, 2025 Week No: 2

DATE	ACTIVITIES UNDERTAKEN
10/06/2025	Simulation of basic digital building blocks using Xilinx Vivado. Design and implementation of sequential components using system verilog.
10/07/2025	Design and implementation of sequential circuit that performs addition based on certain conditions.
10/08/2025	Introduction to simulation using testbenches.
10/09/2025	Mini project to solidify fundamentals - Worked on an algorithm to display and the greatest and second greatest number among three numbers.
10/10/2025	Continued working on the mini project thus writing the algorithm, designing datapath and control word, and controller design.

Name of Supervisor: NERO NOVOR
Title: LAB TECHNICIAN

Date: 8TH FEBRUARY, 2026. Signature:

COLLEGE OF ENGINEERING

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY

Office of the Provost

University Post Office
Kumasi-Ghana
West Africa
Direct Line: 233-3220-60240
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Email: provost.coe@knust.edu.gh

Our Ref: CoE-VACT/2024 Date:

2024/ 2025 Vacation Training Log Sheet

Name of Student: POBI KWAKU GABRIEL Index No: 1854722
Department: ELECTRICAL/ELECTRONICS ENG.
Company: II MICROELECTRONICS LAB
Training Period: From OCTOBER 1, 2025 to NOVEMBER 21, 2025.
Week Begins: OCTOBER 13, 2025 Week Ends: OCTOBER 17, 2025 Week No: 3

DATE	ACTIVITIES UNDERTAKEN
10/12/2025	Mini project presentation of the algorithm to calculate and display the greatest and second greatest number, datapath and controller. ~~and testbench~~
10/14/2025	Programmed the datapath and controller and write testbench for simulation to test different set of values.
10/15/2025	Presented on the software design of the digital system to display the largest and second largest number.
10/16/2025	Designed and simulated memory array with system Verilog. Introduction to digital system application areas like cryptography, systolic arrays.
10/17/2025	Introduction to cryptography and some basic ciphers like playfair, ceasar ciphers. Introduction to Galois Fields.

Name of Supervisor: NERO NUNOR
Title: LAB TECHNICIAN

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Our Ref: CoE-VACT/2024 Date:

2024/ 2025 Vacation Training Log Sheet

Name of Student: POBI KWAKU GABRIEL Index No: 1859722
Department: ELECTRICAL/ELECTRONICS ENG.
Company: TI MICROELECTRONICS LAB
Training Period: From OCTOBER 1, 2025 to NOVEMBER 21, 2025
Week Begins: OCTOBER 20, 2025 Week Ends: OCTOBER 24, 2025 Week No: 4

DATE	ACTIVITIES UNDERTAKEN
10/20/2025	Selected a project in the area of error correction and detection; Designing a BCH encoder and decoder for error detection and correction
10/21/2025	Started studies on the prerequisites for the design - Groups and Finite Fields.
10/22/2025	Continued studies on elements of modern algebra needed for the project - Polynomials over Galois Fields GF(2) and construction of Galois Field $GF(2^m)$.
10/23/2025	Learnt the basic properties and structures of Galois Field $GF(2^m)$ and vector spaces over Finite Fields.
10/24/2025	Presented on the elements of modern algebra - groups, finite fields, polynomials, vector spaces and matrices over finite fields.

Name of Supervisor: NERO NOVOR
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2024/ 2025 Vacation Training Log Sheet

Name of Student: POBI KWAKU GABRIEL Index No: 1859722
Department: ELECTRICAL/ELECTRONIC ENG.
Company: 11 MICROELECTRONICS LAB
Training Period: From OCTOBER 1, 2025 to NOVEMBER 21, 2025
Week Begins: OCTOBER 27, 2025 Week Ends: OCTOBER 31, 2025 Week No: 5

DATE	ACTIVITIES UNDERTAKEN
10/27/2025	Studied the introduction to linear block codes; definitions, generator and parity check matrices as well as systematic linear block codes.
10/28/2025	Learnt how error detection with linear block codes and syndrome and error patterns are done.
10/29/2025	Studied the weight distribution and probability of undetected error and minimum distance of linear block codes.
10/30/2025	Studied the decoding of linear block codes and standard array for decoding linear block codes.
10/31/2025	Presented on linear block codes.

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2024/ 2025 Vacation Training Log Sheet

Name of Student: POBI KWAKU GABRIEL Index No: 1859722
Department: ELECTRICAL/ELECTRONICS ENG
Company: TI MICROELECTRONICS LAB
Training Period: From OCTOBER 1, 2025 to NOVEMBER 21, 2025
Week Begins: NOVEMBER 3, 2025 Week Ends: NOVEMBER 7, 2025 Week No: 6

DATE	ACTIVITIES UNDERTAKEN
11/03/2025	Studied the fundamentals of binary cyclic codes; characterization and structural properties of cyclic codes.
11/04/2025	Learnt about the generator and parity-check matrices of binary cyclic codes and encoding of cyclic codes in systematic form.
11/05/2025	Studied how to perform syndrome calculation and error detection and learnt how to design a general decoder for cyclic codes.
11/06/2025	Studied some special type of decoding; error trapping decoding for cyclic codes, learnt shortened cyclic codes, hamming codes and quadratic residue codes.
11/07/2025	Presented on binary cyclic codes; concepts like generator and parity check matrix, encoding, syndrome calculation and decoding.

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Our Ref: CoE-VACT/2024 Date:

2024/ 2025 Vacation Training Log Sheet

Name of Student: POBI KWAKU GABRIEL Index No: 1859722
Department: ELECTRICAL ELETRONICS ENG
Company: COMPUTER ENGINEERING LAB (IT MICROELECTRONICS LAB).
Training Period: From OCTOBER 1, 2025 to NOVEMBER 21, 2025
Week Begins: NOVEMBER 10, 2025 Week Ends: NOVEMBER 14, 2025 Week No: 7

DATE	ACTIVITIES UNDERTAKEN
11/10/2025	Introduced to the concept of BCH codes; primitive BCH codes, and structural properties of BCH codes.
11/11/2025	Studied syndrome calculation and error detection of BCH codes. Learnt about syndromes and error patterns in BCH codes.
11/12/2025	Learnt about a procedure for decoding BCH codes - Berlekamp-Massey Iterative Algorithm, and also how to find error locations and correct them - Chien's search.
11/13/2025	Studied non-primitive BCH codes.
11/14/2025	Presented on the concepts learnt under BCH codes needed for the project.

Name of Supervisor: NERO NOVOR
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Our Ref: CoE-VACT/2024 Date:

2024/ 2025 Vacation Training Log Sheet

Name of Student: POBI KINAKU GABRIEL Index No: 1859722
Department: ELECTRICAL/ELECTRONIC ENG
Company: COMPUTER ENGINEERING LAB(JI MICROELECTRONICS LAB).
Training Period: From OCTOBER 1, 2025 to NOVEMBER 21, 2025
Week Begins: NOVEMBER 17, 2025 Week Ends: NOVEMBER 21, 2025 Week No: 8

DATE	ACTIVITIES UNDERTAKEN
11/17/2025	Selected a specific example of BCH code; (31, 11, 5) BCH code. Calculated the generator polynomial for this BCH code.
11/18/2025	Designed the encoding circuit using the calculated generator polynomial and assumed a message polynomial to generate a corresponding codeword
11/19/2025	Presented on the encoding of the (31, 11, 5) BCH code.
11/20/2025	Designed a (31, 11, 5) BCH code decoder and implemented with system verilog.
11/21/2025	Presented on the decoder for the (31, 11, 5) BCH code and some limitations and applications of the BCH code.

Name of Supervisor: NERO NOVOR
Title: LAB TECHNICIAN

Date: 18TH FEBRUARY, 2026 Signature:

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Our Ref: CoE-VACT/2025 Date:

2024/ 2025 Vacation Training Appraisal Form

Name of Student: POBI KWAKU GABRIEL Index No: 1859722
Department: ELECTRICAL/ELECTRONIC ENGINEERING
Company: COMPUTER ENGINEERING LAB (II MICROELECTRONICS LAB).
Training Period: From OCTOBER 1, 2025 to NOVEMBER 21, 2025.

Please indicate the appropriate letter applicable to the Student Trainee during the period of the training.
A-Excellent; B-Very Good; C-Good; D-Average; E-Poor

Event	Grade					
	A	B	C	D	E	Cannot be assessed
Punctuality at work		✓				
Ability on job/attitude to work	✓					
Ability to follow instructions	✓					
Relationship with superiors	✓					
Relationship with Colleagues	✓					
Co-operation	✓					
Safety consciousness	✓					
Resourcefulness		✓				
Initiative	✓					
Leadership drive		✓				
General Comments:						

Name of Supervisor: NERO NOVOR
Title: LAB TECHNICIAN
Date: 18TH FEBRUARY 2026
Signature:

For purpose of assessment of the student, the College of Engineering, KNUST will be grateful if this form is filled and either:

1. Posted to The Provost, College of Engineering-KNUST, Kumasi
2. Sealed and given to the Student